

BTC Algorithmic Trading

Professional Teaching Guide: Binance Candle Data, Financial Models, AI Prompting, and Pulse/Tasks Workflows

Prepared for trading coaches teaching beginner-to-intermediate algorithmic market research

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Educational framework - not a live-trading signal service

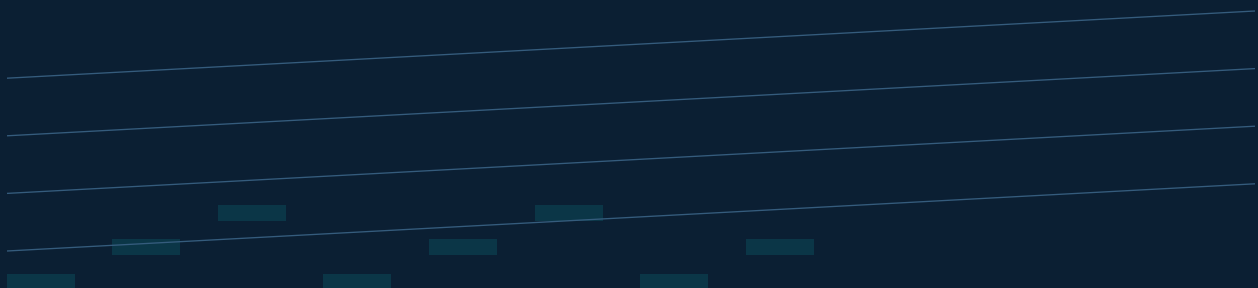


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How to use this PDF

Teach the pipeline in order: clean data first, then features and regimes, then rule-based strategies, then backtesting, then AI as a research assistant. Do not allow AI narratives to replace objective entry, exit, stop, and sizing rules.

Quick-start instructor map

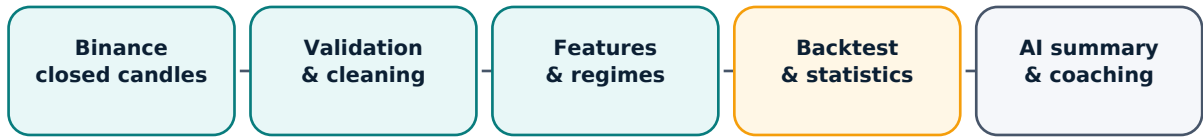
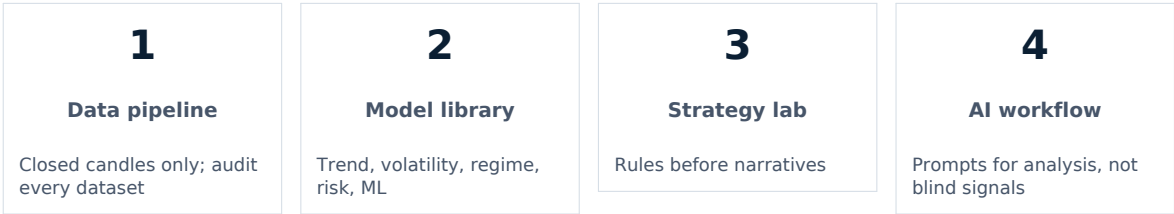
Stage	Instructor action	Student artifact
1	Pull closed BTCUSDT candles and audit timestamps.	Clean candle file plus data-quality notes.
2	Build features slowly: returns, ATR, EMA, volume, regime.	Feature table with formulas and caveats.
3	Select one strategy template and write objective rules.	Entry/exit/stop/sizing sheet.
4	Backtest with fees, slippage, and out-of-sample periods.	Performance and risk memo.
5	Use AI prompts to explain, critique, and schedule review tasks.	AI model brief and weekly learning plan.

Do	Do not
Use AI to summarize evidence, detect missing assumptions, and generate review questions.	Use AI as a blind buy/sell signal engine.
Teach model failure modes and no-trade rules early.	Let the student optimize indicators before defining risk.

TEACHING INTENT

1. Executive Summary

This guide turns BTC candle analysis into a professional, teachable workflow. The central lesson is that algorithmic trading is not a prediction contest; it is a disciplined process of data validation, model selection, backtesting, statistical review, risk control, and repeated learning.



Core principle: AI summarizes evidence and checks assumptions; the trading system defines the rules and risk.

Core rule for the student

AI may summarize evidence, test assumptions, and explain model output. It should not be treated as an oracle telling the student to buy or sell BTC. The system must define valid setup, stop, target, risk, and no-trade conditions before any model output is considered.

Recommended teaching philosophy

- Start with closed OHLCV candles and timestamp integrity before any indicator or AI model.
- Use simple rule-based strategies before machine learning.
- Evaluate models by expectancy, drawdown, robustness, and regime fit - not by one profitable chart example.
- Use prompts as reusable research checklists so the student learns process, not prediction dependency.
- For a High D / High C student, require hard rules: no impulsive discretionary entries, no overfitting perfectionism, and no live deployment before paper trading.

FROM RAW CANDLES TO TEACHABLE DECISIONS

2. Professional Workflow Architecture

The workflow below separates data engineering, research, strategy design, risk management, and AI-assisted interpretation. This separation is crucial because most beginner algorithmic traders mix these tasks together and mistake a persuasive narrative for a tested edge.

Layer	Purpose	Student deliverable	Common failure
Data ingestion	Pull BTCUSDT candles from Binance REST or WebSocket sources.	Clean CSV or database table with closed candles only.	Using live, unfinished candles or missing timestamps.
Data validation	Check duplicates, gaps, timezone, numeric fields, abnormal spikes.	Data audit report with pass/fail status.	Modeling dirty data and trusting false indicators.
Feature engineering	Calculate returns, volatility, trend, momentum, volume, candle structure.	Feature table with formulas and caveats.	Adding too many features before understanding them.
Regime detection	Label bull, bear, chop, compression, expansion.	Regime map and frequency distribution.	Applying trend systems during chop.
Strategy rules	Define objective entry, exit, stop, target, sizing, no-trade filters.	Code-ready rule sheet.	Using indicators without a complete trading system.
Backtest & stats	Evaluate fees, slippage, drawdown, expectancy, walk-forward results.	Performance review and model risk memo.	Cherry-picked periods and lookahead bias.
AI interpretation	Summarize evidence, explain results, generate research questions.	AI trend brief, coaching notes, task prompts.	Asking AI for blind buy/sell predictions.

Minimum viable research stack

Component	Beginner choice	Professional upgrade
Data	Binance BTCUSDT 1h and 4h klines	Database with 1m-1d candles and live WebSocket feed
Storage	CSV files	SQLite/PostgreSQL plus data-quality logs
Research	Python/pandas notebook	Versioned notebooks, unit tests, reproducible runs
Backtesting	Vectorized rule test	Event-driven backtester with fees, slippage, latency
Modeling	Indicators and statistical tests	Walk-forward ML/regime models with monitoring
AI	Prompt-based analysis	Scheduled model-health summaries and structured reports

DATA FOUNDATION

3. Binance BTC Candle Data Pipeline

BTC analysis begins with reliable candles. Binance Spot kline/candlestick data is available through the REST endpoint GET /api/v3/klines; the endpoint returns open time, OHLC prices, volume, close time, quote volume, number of trades, taker-buy volumes, and an unused field. Klines are identified by open time, and the documented request limit is 1,000 candles for this endpoint. [1]

For live monitoring, Binance kline WebSocket streams use a symbol and interval stream name. The kline payload includes a boolean field indicating whether the candle is closed. Models and backtests should normally use only closed candles, because still-forming candles can produce lookahead-style errors and unstable signals. [2]

Field	Meaning	Modeling note
open_time	Candle open timestamp	Primary key; verify no duplicates or gaps.
open, high, low, close	OHLC prices	Convert strings to numeric; check high >= max(open, close) and low <= min(open, close).
volume	Base asset volume	Useful for liquidity and participation.
close_time	Candle close timestamp	Use to confirm closed intervals and align forward returns.
quote_asset_volume	Quote currency volume	Often better than base volume for dollar activity.
number_of_trades	Trades inside candle	Activity intensity; useful as a volume confirmation feature.
taker_buy_base_volume	Aggressive buy base volume	Can estimate buy-pressure ratio.
taker_buy_quote_volume	Aggressive buy quote volume	Useful for notional pressure.

Closed candle discipline

When using WebSocket data, ignore model signals from candles where the close flag is false. Students should learn that a beautiful mid-candle breakout can disappear by the candle close.

Data audit checklist

- Timestamps are sorted ascending and unique.
- Expected interval spacing is consistent; missing candles are counted and repaired or flagged.
- OHLC fields are numeric and internally consistent.
- Zero-volume candles are reviewed instead of silently dropped.
- Timezone handling is explicit: UTC for storage, local time only for presentation.
- The last candle is excluded unless the data source confirms it is closed.
- Training, validation, and test periods are separated chronologically.

MODEL SELECTION

4. Financial and Quantitative Model Library

The model library is organized by purpose. Some models estimate trend, others classify regime, others evaluate risk, and derivatives models such as Black-Scholes or finite-difference methods are mainly for BTC options rather than ordinary spot-candle direction prediction.

Model family	Best use with BTC	What it teaches	Primary caveat
Indicators	Trend, momentum, volatility, volume context	How rules convert raw candles into interpretable features	Indicators are not strategies by themselves.
Return/volatility models	Risk scaling, regime detection, volatility forecasts	BTC risk changes over time	Volatility does not equal direction.
Statistical tests	Validate signal forward returns and win rates	Difference between observed edge and random luck	Small samples mislead.
Regime models	Trend/chop/compression/expansion labels	A strategy must match market condition	Regimes can shift quickly.
Time-series forecasting	Baseline forecasts and scenario exploration	Forecast uncertainty and validation	Price-level forecasting is often fragile.
Machine learning	Probability of continuation/reversal/chop	Feature engineering and overfitting control	Requires walk-forward validation.
Monte Carlo	Equity curve and drawdown simulation	One backtest is not enough	Simulation assumptions matter.
Derivatives models	BTC options pricing and volatility	Delta, gamma, vega, implied volatility	Needs options data, not just spot candles.

Teaching sequence

Do not begin with LSTMs, transformers, Black-Scholes, or exotic derivatives. Begin with returns, volatility, ATR, moving averages, regime labels, and one rule-based strategy that can be backtested honestly.

4.1 Foundation indicators

Model	Use	Rule example	Teaching point
EMA / SMA	Trend direction and smoothing	Trade long only when close > 200 EMA; short only when close < 200 EMA.	All moving averages lag; they filter but do not predict.
VWAP	Intraday fair value and pullback zone	In trend, enter continuation after VWAP retest and rejection.	Works best when session structure matters.
RSI	Momentum exhaustion	Use RSI < 30 only in range regime; avoid counter-trend in strong trends.	Oversold can become more oversold.
MACD	Momentum acceleration /deceleration	Use histogram expansion as confirmation, not primary signal.	Late signals in fast crypto moves.
Bollinger Bands	Volatility expansion/compression	Breakout when band width leaves low percentile and price closes outside range.	Band touch alone is not a trade.
ATR	Stop distance and risk sizing	Stop = 1.5 ATR; position size based on max dollar risk.	ATR measures movement, not direction.

4.2 Core formula cards

Concept	Formula / rule	Use with BTC
Log return	$r_t = \ln(C_t / C_{t-1})$	Standardize returns; compare timeframes; build labels.
Rolling volatility	$\sigma_N = \text{stdev}(r \text{ over } N \text{ candles})$	Detect high/low volatility regimes and scale risk.
EWMA volatility	Recent returns receive greater weight than older returns.	React faster to BTC volatility expansion.
ATR	Average of true range over N candles.	Stops, targets, volatility-adjusted position sizing.
Two-tailed t-test	H0: mean forward return = 0; H1: mean forward return $\neq$ 0.	Test whether a signal's average forward return differs from zero.
Binomial test	H0: win rate equals baseline.	Test whether a signal's win rate is unusual.
Bootstrap	Resample observed trades/returns many times.	Estimate confidence intervals without assuming normality.
VaR / CVaR	VaR = percentile loss; CVaR = average tail loss beyond VaR.	Measure tail risk and stress losses.
Kelly fraction	$f^* = (bp - q) / b$	Position sizing theory; use fractional Kelly only.

Instructor caution

A statistically significant result can still be untradable after fees, slippage, drawdown, and execution constraints. Make the student separate statistical significance from practical tradability.

4.3 Advanced model families

Model	What it does	Best BTC use	Not ideal for
GARCH	Models volatility clustering.	Forecast next-period volatility and risk scaling.	Directional buy/sell decisions by itself.
Hurst exponent	Measures trending vs mean-reverting tendency.	Regime filter for trend or mean-reversion systems.	Single-candle timing.
Hidden Markov Model	Infers unobserved regimes from returns/volatility.	Label bull/bear/chop/high-vol regimes.	Small data or constantly changing feature definitions.
Kalman filter	Smooths noisy signals and estimates dynamic state.	Adaptive trend estimate on noisy BTC data.	A standalone edge.
Change-point detection	Finds structural breaks in behavior.	Detect volatility or trend regime shifts.	Predicting exact reversal prices.
ARIMA/ARIMAX	Classical time-series forecasting.	Educational baseline and feature-added forecasting.	High-frequency price prediction without regime context.
Random forest / XGBoost	Nonlinear tabular classification.	Predict probability of continuation/chop using engineered features.	Unvalidated live deployment.
LSTM / transformer	Sequence modeling on time series.	Advanced experiments with large, clean datasets.	Beginners; easy to overfit and hard to explain.

4.4 Black-Scholes, finite-difference, and BTC options

Black-Scholes and finite-difference methods are valuable in a BTC curriculum, but the instructor should clearly position them as derivatives and volatility tools, not as ordinary BTC spot-candle trend predictors.

Model	Inputs	Use	Teaching caveat
Black-Scholes	Spot price, strike, time to expiry, risk-free rate, volatility	Theoretical call/put pricing; Greeks such as delta, gamma, theta, vega.	Requires options data and assumptions; not a direct BTC spot signal.
Implied volatility	Observed option price plus option parameters	Market-implied expectation of future volatility.	IV can stay high/low and does not specify direction.
Finite-difference PDE	Grid over price and time plus volatility/rate assumptions	Numerical option pricing, barriers, early exercise variants.	Advanced; best after student understands options payoff diagrams.
Monte Carlo path pricing	Simulated price paths and payoff function	Option valuation, scenario analysis, strategy drawdown simulation.	Outputs depend on path assumptions.

Black-Scholes call concept:
 $C = S * N(d1) - K * exp(-rT) * N(d2)$

$d1 = [ln(S/K) + (r + 0.5*sigma^2)T] / [sigma*sqrt(T)]$
 $d2 = d1 - sigma*sqrt(T)$

Teaching interpretation:

- S is current BTC price.
- K is option strike.
- sigma is volatility.
- The model explains option value and Greeks, not candle direction.

Recommended placement

Teach Black-Scholes after the student already understands returns, volatility, distributions, and risk. Otherwise they may confuse option-pricing theory with spot-trading prediction.

FROM MODELS TO RULES

5. BTC Strategy Templates for Teaching

Each strategy template below is intentionally rule-based. The student may later use AI to summarize market evidence, but the strategy must be expressible without vague language.

Template	Best regime	Entry	Exit / risk rule
Trend-following breakout	High-volatility bull/bear trend	Close breaks above/below N-candle range in direction of higher-timeframe EMA bias.	Stop beyond range or 1.5 ATR; target 2R or trailing stop.
VWAP pullback continuation	Directional intraday trend	Price above/below VWAP and 50 EMA; pullback rejects VWAP zone.	Stop behind pullback structure; target prior high/low or 1.5R-2R.
Volatility expansion breakout	Compression into expansion	Bollinger width in low percentile; close outside compression range with volume z-score confirmation.	Stop inside range; trail with ATR or exit on failed close back inside.

Template	Best regime	Entry	Exit / risk rule
Mean reversion in chop	Low ADX / range-bound market	RSI extreme near range boundary; no higher-timeframe trend conflict.	Stop outside range; target VWAP or range midpoint.

Strategy design worksheet

Question	Required answer before testing
What regime is this for?	Trend, chop, compression, expansion, high volatility, or low volatility.
What is the exact entry?	A rule that can be coded with no visual interpretation.
What invalidates the trade?	Stop price, time stop, or signal failure rule.
How is position size calculated?	Dollar risk / stop distance; never fixed lot size alone.
What fees/slippage are assumed?	Explicit maker/taker fee and slippage estimate.
What would make the system pause?	Daily loss, volatility shock, missing data, regime mismatch.

PROOF BEFORE CONFIDENCE

6. Backtesting, Statistics, and Model Validation

A backtest is a research artifact, not proof of future profit. Teach the student to treat every backtest as a hypothesis that must survive fees, slippage, out-of-sample testing, sensitivity analysis, and Monte Carlo stress testing.

Metric	What it measures	Healthy interpretation	Warning sign
Expectancy	Average profit/loss per trade	Positive after costs across regimes.	Positive only before fees or in one period.
Profit factor	Gross profit / gross loss	Above 1 after realistic costs.	High value with tiny sample size.
Max drawdown	Worst peak-to-trough equity loss	Acceptable relative to account and psychology.	Strategy makes money but drawdown is untradeable.
Sharpe / Sortino	Risk-adjusted return	Useful for comparison, not standalone proof.	Inflated by smooth backtest assumptions.
Win rate	Percent of profitable trades	Interpreted with avg win/loss.	High win rate with rare catastrophic losses.
Average R	Return normalized by initial risk	Makes results comparable across sizes.	Not tracked, so risk is unclear.
Walk-forward	Robustness across time	Parameters hold up in unseen periods.	Optimized on full data.

Validation standard

A strategy is not ready for paper trading until it has clean data, explicit rules, realistic costs, out-of-sample review, regime breakdown, and drawdown simulation. A strategy is not ready for live trading until it has survived paper trading without rule violations.

Monte Carlo stress test

Monte Carlo is one of the most valuable teaching tools because it shows that even a positive-expectancy strategy can have painful losing streaks. Simulate many possible equity curves using the observed trade distribution, or resample the sequence of R-multiples.

Output	Teaching question
Median ending equity	What is the central expectation?
5th percentile ending equity	What happens in a bad but plausible run?
Probability of loss	How often does the strategy lose money in simulation?
Probability of 10/20/30% drawdown	Can the student psychologically and financially tolerate this?
Longest losing streak	Will the student abandon the system at the worst time?

PROBABILITY, NOT CERTAINTY

7. AI and Machine Learning Model Design

For BTC, beginner ML should usually classify probabilities of regime, continuation, reversal, or no-trade conditions rather than predicting an exact future price. This makes the model easier to validate and harder to misuse.

Design choice	Recommended beginner implementation	Professional upgrade
Target label	Next N-candle return > X ATR, < -X ATR, or neutral.	Triple-barrier labeling with stop, target, and time horizon.
Features	Returns, ATR, EMA slope, RSI, ADX, volume z-score, candle body/wick ratios.	Multi-timeframe features, order-flow proxies, external macro/liquidity features.
Baseline	Logistic regression or decision tree.	Gradient boosting, calibrated probabilities, ensemble models.
Validation	Chronological train/test split.	Walk-forward and purged cross-validation.
Metrics	Precision/recall, confusion matrix, calibration, strategy expectancy.	Profit-aware metrics and regime-specific deployment monitoring.
Interpretability	Feature importance and model explanation.	SHAP/partial dependence plus drift monitoring.

Modeling principle

The model should produce probabilities such as 57% bullish continuation, 28% chop, 15% bearish continuation. The trading system then decides whether that probability is strong enough after costs and risk rules.

Overfitting checklist

- Too many indicators or optimized parameters for the sample size.
- No walk-forward validation or out-of-sample holdout.
- Ignoring fees, slippage, spreads, latency, and partial candles.
- Testing many signals and only reporting the best one.
- Changing rules after seeing failed examples without revalidating out-of-sample.
- Using future data accidentally in features or labels.

REUSABLE AI RESEARCH PROMPTS

8. ChatGPT Prompt Library for BTC Data Modeling

The following prompts are designed to be pasted into ChatGPT with a CSV, model output, backtest summary, or written strategy description. They force structure, identify assumptions, and keep the student focused on evidence.

Prompting rule

Every prompt should ask ChatGPT to state assumptions, separate evidence from inference, flag data-quality issues, and avoid trade recommendations unless the exercise is explicitly a simulated strategy-design assignment.

Prompt 1 - Data Cleaning and Audit

Objective

Check whether Binance candle data is safe to model.

You are a quantitative trading analyst.

I am uploading Binance BTCUSDT candle data with these columns:
open_time, open, high, low, close, volume, close_time, quote_asset_volume,
number_of_trades, taker_buy_base_volume, taker_buy_quote_volume.

Audit the dataset before modeling.

Check for:

- 1. Missing candles
- 2. Duplicate timestamps
- 3. Incorrect timestamp order
- 4. Non-numeric OHLCV fields
- 5. Zero-volume candles
- 6. Abnormal price spikes
- 7. Timezone assumptions
- 8. Whether candles are closed or still forming

Return:

- Data quality summary
- Problems found
- Fixes needed
- Cleaned schema recommendation
- Whether the data is safe for backtesting

Prompt 2 - Feature Engineering

Objective

Convert raw candles into trend, volatility, volume, and structure features.

You are a quant researcher building features for BTCUSDT candle data.

Using the uploaded Binance candle data, propose a feature set for algorithmic trend analysis.

Include:

1. Return features
2. Volatility features
3. Trend features
4. Momentum features
5. Volume features
6. Candle-structure features
7. Taker-buy pressure features
8. Multi-timeframe features

For each feature, explain:

- Formula
- Why it matters
- Whether it is leading, lagging, or contextual
- Risk of overfitting

Return the final feature table.

Prompt 3 - Trend Regime Classification

Objective

Label each period as trend, chop, compression, or expansion.

You are an algorithmic trading coach.

Analyze BTCUSDT candle data and classify each period into one of these regimes:

1. Bull trend
2. Bear trend
3. Sideways chop
4. High-volatility expansion
5. Low-volatility compression

Use features such as:

- EMA slope
- Price versus 50 EMA and 200 EMA
- ATR percentile
- Bollinger Band width
- ADX
- Volume z-score
- Recent return distribution

Return:

- Regime definitions
- Labeling rules
- Summary of how often each regime occurs
- Which regimes suit trend-following
- Which regimes suit mean reversion
- Which regimes should be avoided

Prompt 4 - Statistical Edge Test

Objective

Test whether a signal has meaningful forward returns.

You are a financial statistician.

I want to test whether this BTC signal has a statistically significant edge:

Signal:
[INSERT SIGNAL RULE]

Test whether the forward return after the signal is significantly different from zero.

Use:

- Forward returns over 1, 3, 5, 10, and 20 candles
- Two-tailed t-test
- Bootstrap confidence interval
- Win-rate analysis
- Average win/loss
- Profit factor
- Maximum adverse excursion if possible

Return:

- Null hypothesis
- Alternative hypothesis
- Test results
- p-values
- Confidence intervals
- Practical interpretation
- Whether the edge is tradable after fees and slippage

Prompt 5 - Strategy Builder

Objective Turn a concept into rules that can be coded.

You are a professional systematic trading strategist.

Using BTCUSDT candle data, design a rule-based strategy based on this concept:

Strategy concept:
[INSERT: breakout / VWAP pullback / mean reversion / volatility expansion]

Requirements:

1. Objective entry rule
2. Objective exit rule
3. Stop-loss rule
4. Take-profit rule
5. Position-sizing rule
6. Market-regime filter
7. Maximum daily loss rule
8. No-trade conditions
9. Fees and slippage assumptions
10. Backtest metrics to evaluate

Do not give vague trading advice. Return a precise rule set that can be coded.

Prompt 6 - Backtest Review

Objective Evaluate whether a strategy result is robust or fragile.

You are an institutional trading performance analyst.

I am uploading backtest results for a BTCUSDT algorithmic strategy.

Evaluate:

1. Net profit
2. Profit factor
3. Expectancy
4. Win rate
5. Average win versus average loss
6. Max drawdown
7. Sharpe ratio
8. Sortino ratio
9. Trade frequency
10. Exposure time
11. Performance by regime
12. Performance by time of day
13. Sensitivity to fees and slippage
14. Signs of overfitting

Return:

- Executive summary
- Strengths
- Weaknesses
- Hidden risks
- Whether the strategy is robust
- What must improve before paper trading

Prompt 7 - Monte Carlo Risk Simulation

Objective Translate backtest results into risk and drawdown probabilities.

You are a risk manager.

Using the strategy results I upload, run or design a Monte Carlo simulation.

Inputs:

- Win rate
- Average win
- Average loss
- Trade count
- Max drawdown
- Profit factor
- Starting capital
- Risk per trade

Simulate 10,000 possible equity curves.

Return:

1. Median ending equity
2. 5th percentile ending equity
3. 95th percentile ending equity
4. Probability of losing money
5. Probability of 10%, 20%, and 30% drawdown
6. Longest likely losing streak
7. Recommended max risk per trade
8. Whether the strategy is psychologically tradable

Prompt 8 - Machine Learning Design

Objective Design a probability model without overfitting.

You are a machine-learning researcher specializing in financial time series.

I want to build an ML model using BTCUSDT candle data.

Goal:

Predict whether BTC will move at least [X] ATR in either direction over the next [N] candles.

Please design:

1. Target label
2. Feature set
3. Train/validation/test split
4. Walk-forward validation method
5. Baseline model
6. Advanced model candidates
7. Metrics to use
8. Overfitting checks
9. Feature importance analysis
10. Deployment warning checklist

Important: do not predict price directly unless justified. Prefer probability outputs.

Prompt 9 - AI Trend Summary

Objective Summarize market evidence without issuing a blind signal.

You are an AI trading analyst.

Using the uploaded BTCUSDT candle data, summarize the current market condition.

Analyze:

1. Multi-timeframe trend
2. Volatility regime
3. Momentum
4. Volume confirmation
5. Key support/resistance from recent candles
6. Whether conditions favor trend-following, breakout, mean reversion, or no trade
7. What evidence would invalidate the current read

Return:

- Bullish evidence
- Bearish evidence
- Neutral/chop evidence
- Best-fit strategy type
- Risk warnings
- Confidence level from 0 to 100

Do not provide a trade instruction. Provide model evidence only.

Prompt 10 - Student-Friendly Explanation

Objective Translate model output into plain English and quiz the student.

You are teaching a beginner algorithmic trading student.

Explain the results of this BTC model in plain English.

Avoid jargon unless you define it.

Cover:

1. What the model is trying to detect
2. What data it uses
3. What it is good at
4. What it is bad at
5. Why it can fail
6. Why backtesting is not live trading
7. What the student should check before trusting it

End with five quiz questions.

AI OPERATING CADENCE

9. Pulse and Scheduled Task Workflow

ChatGPT Tasks can run automated prompts later on a one-time or recurring schedule and notify the user when complete. Tasks are supported on several ChatGPT platforms; OpenAI's help documentation notes a current limit of 10 active tasks and that file uploads are not supported inside Tasks. [3]

Pulse is a Pro feature that performs asynchronous research once per day and delivers visual summaries; OpenAI's Pulse documentation describes availability on Web, iOS, and Android, with memory required. [4]

Workflow implication

Because scheduled Tasks do not support file uploads, use an external script or notebook to pull Binance data and produce a compact text/CSV summary. Then use Tasks/Pulse prompts for daily research briefs, model-health reminders, teaching plans, and review workflows.

Workflow	Schedule	Purpose
Daily BTC Market Structure Brief	Every day, 8:00 AM ET	Frame the market for teaching without issuing trade recommendations.
Daily Model Health Check	After data pipeline update	Confirm data freshness, missing candles, regime alignment, and signal validity.
Weekly Strategy Review	Friday, 5:00 PM ET	Review theoretical signals, regimes, false-signal causes, and student lessons.
Monday Lesson Plan	Monday, 7:30 AM ET	Create a one-hour lesson plan based on the week's topic.
Overfitting Warning Review	Wednesday, noon ET	Force a skeptical model-risk review.
Risk Drill	Tuesday/Thursday, 9:00 AM ET	Practice position sizing and max-loss calculations.
Monte Carlo Friday	Friday, 3:00 PM ET	Teach drawdown, equity-curve variation, and risk-of-ruin concepts.
Monthly Model Improvement Review	First day of month	Decide what to keep, remove, improve, and test next.

Task 1 - Daily BTC Market Structure Brief

Objective Suggested schedule: Daily at 8:00 AM ET.

Give me a daily BTCUSDT algorithmic trading education brief.

Focus on:

1. Overnight BTC market structure
2. Current trend regime: bull, bear, chop, volatility expansion, or compression
3. Whether today is better suited for breakout, trend-following, mean reversion, or no-trade education
4. Key volatility conditions
5. One concept to teach my student today
6. One risk-management reminder
7. One question I should ask the student

Do not give a trade recommendation. Frame this as education and model evaluation.

Task 2 - Daily Model Health Check

Objective Suggested schedule: After pipeline update.

Review my BTCUSDT model workflow and produce a model-health checklist.

Check:

1. Was fresh candle data collected?
2. Are there missing candles?
3. Did volatility regime change?
4. Did the current market regime match the model's preferred regime?
5. Did any signal fire?
6. Was the signal inside or outside the model's tested conditions?
7. What should be logged for future backtesting?

Return:

- Green/yellow/red model status
- Main concern
- What to review before trusting any output

Task 3 - Weekly Strategy Review

Objective Suggested schedule: Friday at 5:00 PM ET.

Create a weekly BTC strategy review for my algorithmic trading student.

Summarize:

1. Number of model signals this week
2. Which regimes appeared most often
3. Which strategy type performed best in theory: breakout, trend-following, VWAP pullback, or mean reversion
4. What conditions caused false signals
5. Whether volatility increased or decreased
6. What the student should learn from this week

End with:

- Three coaching notes
- Three homework tasks
- One backtesting improvement idea

Task 4 - Monday Lesson Plan

Objective

Suggested schedule: Monday at 7:30 AM ET.

Build a one-hour lesson plan for teaching BTC algorithmic trading this week.

Student level: beginner to intermediate.

Include:

1. Topic of the week
2. Learning objective
3. Simple explanation
4. BTC candle example to demonstrate
5. Model or indicator to teach
6. Coding/backtesting assignment
7. Common beginner mistake
8. Short quiz

Prioritize practical trading research, not hype or prediction.

Task 5 - Overfitting Warning Review

Objective

Suggested schedule: Wednesday at noon ET.

Act as a skeptical quant reviewer.

Review the BTC strategy/model I am currently teaching and identify overfitting risks.

Look for:

1. Too many indicators
2. Too many optimized parameters
3. Weak sample size
4. No walk-forward test
5. No fees/slippage
6. Poor out-of-sample results
7. Unrealistic trade execution
8. Lookahead bias
9. Survivorship/data bias
10. Cherry-picked time periods

Return:

- Top five overfitting risks
- Severity score for each
- How to fix each one
- One simplified version of the strategy

Task 6 - Risk Management Drill

Objective

Suggested schedule: Tuesday and Thursday at 9:00 AM ET.

Create a BTC algorithmic trading risk-management drill for my student.

Use this format:

1. Scenario
2. Account size
3. Strategy signal
4. Stop distance
5. Volatility condition
6. Student must calculate position size
7. Student must calculate max loss
8. Student must decide whether the trade is valid

Then provide the answer key and explain the reasoning.

Task 7 - Monte Carlo Friday

Objective

Suggested schedule: Friday at 3:00 PM ET.

Teach one Monte Carlo concept using BTC strategy results.

Cover:

1. What Monte Carlo simulation means
2. Why one backtest is not enough
3. How equity curves can vary
4. How losing streaks happen even in profitable systems
5. How to choose risk per trade
6. What probability of drawdown means

Include a simple numeric example. End with one homework question.

Task 8 - Monthly Model Improvement Review

Objective

Suggested schedule: First day of each month at 9:00 AM ET.

Create a monthly BTC algorithmic trading model-improvement report.

Review:

1. Which model assumptions should be retested
2. Which features may be useless
3. Which regimes caused poor performance
4. Whether model complexity increased too much
5. Whether the strategy still has positive expectancy
6. What should be removed, simplified, or improved
7. What should be tested next month

Return:

- Keep
- Remove
- Improve
- Test next
- Student lesson focus

INSTRUCTOR ROADMAP

10. Eight-Lesson Curriculum Plan

The curriculum sequence below moves from raw candles to AI-assisted modeling without letting the student skip the hard parts of data integrity, rule definition, backtesting, and risk management.

Lesso n	Topic	Learning objective	Assignment
1	Candles and Binance data	Understand OHLCV, timeframes, closed candles, and timestamps.	Pull 1,000 BTCUSDT 1h candles; convert timestamps; plot close.
2	Returns and volatility	Use returns instead of raw price; calculate ATR and rolling vol.	Calculate log returns and identify high-volatility windows.
3	Trend vs chop	Classify regimes with EMA, ADX, Bollinger width, and volatility.	Label each day as trend, chop, or expansion.
4	Rule-based strategy	Define objective entry, exit, stop, target, and sizing.	Build a 20-candle breakout with ATR stop.
5	Backtesting basics	Handle fees, slippage, lookahead bias, train/test split.	Backtest one strategy and report expectancy/drawdown.
6	Statistical testing	Use t-test, bootstrap, and win-rate tests appropriately.	Test signal forward returns at multiple horizons.
7	Monte Carlo risk	Simulate equity curves and losing streaks.	Run 10,000 simulated equity curves from trade R results.
8	AI-assisted modeling	Use prompts for evidence summaries, model design, and review.	Create a model brief and no-trade checklist.

Student behavior guardrails

For students who are aggressive, analytical, or prone to over-optimizing, the instructor should make the following rules part of the course, not optional advice.

- No model may be judged from one chart example.
- No indicator is allowed unless the student can explain what it measures and why it belongs in the strategy.
- No backtest is accepted without fees, slippage, sample period, and out-of-sample validation.
- No AI-generated market read is accepted unless it states invalidation and evidence against the thesis.
- No live deployment until the system has a paper-trading log and rule-violation review.

PROFESSIONAL OPERATING STANDARDS

11. Model Governance and Risk Controls

A professional-grade algorithmic trading curriculum should teach governance early. Governance means the student knows what the model is allowed to do, when it should be ignored, and how it will be monitored.

Control	Rule	Why it matters
Data freshness	No model output if latest closed candle is missing or delayed.	Prevents stale signals.
Closed-candle rule	Backtests and signals use only confirmed closed candles.	Prevents false mid-candle signals.
Regime filter	Strategy only runs in tested regime.	Stops trend systems from trading chop.
Risk cap	Max risk per trade and max daily loss defined in advance.	Prevents emotional escalation.
Model pause	Pause after abnormal volatility, exchange outage, missing data, or model drift.	Avoids trading when assumptions fail.
Version control	Every rule change gets a version number and reason.	Stops hidden curve fitting.
Review cadence	Daily health check, weekly strategy review, monthly improvement review.	Creates continuous learning.

Professional standard

The student should maintain a model card for every strategy: intended regime, features, parameters, sample period, validation method, risk limits, failure modes, and retirement criteria.

Model card template

Field	Prompt to complete
Strategy name	What is this strategy called and what behavior does it exploit?
Instrument/timeframe	Which symbol and timeframe were tested?
Regime	Which market regimes are allowed and prohibited?
Features	Which features are used and why?
Rules	Entry, exit, stop, target, sizing, and no-trade conditions.
Backtest period	Training, validation, and test periods.
Costs	Fees, spread, slippage, latency assumptions.
Risk limits	Max trade risk, daily loss, max open exposure.
Failure modes	When should the model be paused or retired?

REFERENCE

Appendix A - Code-Friendly BTC Data Schema

Column	Type	Description	Validation
open_time	datetime/int64	Candle open time	Unique; sorted ascending; interval gap check.
open	float	Open price	Positive; within high-low.
high	float	High price	High >= open, close, low.
low	float	Low price	Low <= open, close, high.
close	float	Close price	Positive; used for returns and labels.
volume	float	Base volume	Flag zero or extreme values.
close_time	datetime/int64	Candle close time	Should align with interval.
quote_asset_volume	float	Quote volume	Useful for notional activity.
number_of_trades	int	Trade count	Non-negative integer.
taker_buy_base_volum e	float	Aggressive buy base volume	0 <= value <= total volume, usually.
taker_buy_quote_volu me	float	Aggressive buy quote volume	Compare to quote volume.
is_closed	bool	Closed-candle flag for live streams	Must be true for modeling.

Suggested engineered fields

- log_return_1: $\ln(\text{close}_t / \text{close}_{t-1})$
- atr_14: average true range over 14 candles
- ema_50_slope and ema_200_slope
- rsi_14 and macd histogram
- bb_width_percentile
- volume_zscore and trade_count_zscore
- taker_buy_ratio = $\text{taker_buy_base_volume} / \text{volume}$
- body_pct = $\text{abs}(\text{close} - \text{open}) / (\text{high} - \text{low})$
- upper_wick_pct and lower_wick_pct
- forward_return_N and forward_atr_move_N for labels

REFERENCE

Appendix B - Student Glossary

Term	Plain-English definition
Algorithmic trading	Using defined rules and code to research or execute trading decisions.
Backtest	A historical simulation of how a strategy would have performed.
Expectancy	Average amount a strategy makes or loses per trade after costs.
Regime	The current market environment, such as trend, chop, compression, or expansion.
ATR	Average true range; a volatility measure often used for stops and sizing.
Lookahead bias	Using information in a backtest that would not have been known at the time.
Overfitting	Tuning a model so tightly to past data that it fails on new data.
Walk-forward validation	Testing a model on unseen future windows after training on prior windows.
Monte Carlo	Repeated simulation to estimate possible outcome ranges and drawdowns.
VaR / CVaR	Risk measures for likely loss threshold and average extreme-tail loss.
Black-Scholes	A classic option-pricing model, useful for options education and volatility concepts.

Final instructor message

Algorithmic trading education should reward process quality. A student who can reject a weak signal, identify missing data, and explain why a model should not trade is making real progress.

REFERENCES

Appendix C - Source Notes

These sources were consulted for platform-specific details. General descriptions of statistical, risk, and financial models are included as educational background and should be validated in the student's own research environment before use.

Ref	Source	Used for
[1]	Binance Spot API - Market Data Endpoints	Kline REST endpoint, parameters, intervals, response fields, and request limit.
[2]	Binance Spot API - WebSocket Streams	Kline stream naming, update behavior, payload fields, and closed-candle flag.
[3]	OpenAI Help - Tasks in ChatGPT	Task scheduling, notifications, active task limit, and unsupported tools such as file uploads.
[4]	OpenAI Help - ChatGPT Pulse	Pulse availability, daily asynchronous research behavior, and memory requirement.

Disclaimer: This document is for education, research, and paper-trading workflow design. It is not investment advice, financial advice, or a recommendation to trade BTC or any other asset. Trading crypto assets can involve substantial risk, including the loss of principal.